

ALIGNED AMINOS

GROWTH & GH AXIS

PRODUCT DATA SHEET

RESEARCH USE ONLY

GROWTH & GH AXIS

IGF-1 LR3

IGF-1 LR3 – 1 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Long [Arg3] Insulin-like Growth Factor-1
SYNONYMS / ALIASES	IGF-1 LR3; LR3-IGF-1; Long R3 IGF-1
CAS NUMBER	946870-92-4
MOLECULAR FORMULA	C400H625N111O115S9
MOLECULAR WEIGHT	Approximately 9111.5 g/mol
AMINO ACID SEQUENCE	83 residues; native human IGF-1 (70 residues) with an Arg3 substitution (Glu to Arg at position 3 of the IGF-1 portion) and a 13-amino-acid N-terminal extension derived from porcine growth hormone (MFPAMPLSSLFVN)
SOURCE	Recombinant (typically expressed in E. coli)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	1 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14-28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only); acidified water (0.1% acetic acid) may improve solubility
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Contains three intramolecular disulfide bonds critical to folding. Avoid reducing agents and repeated freeze-thaw cycles.

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STRUCTURAL NOTES

IGF-1 LR3 is a recombinant 83-amino-acid analog of human Insulin-like Growth Factor-1 (IGF-1). Two structural modifications differentiate it from native IGF-1: a single-residue substitution (Glu3 to Arg3) in the IGF-1 backbone, and an N-terminal extension of 13 amino acids derived from porcine growth hormone. These modifications reduce binding affinity for the IGF-binding proteins (IGFBPs) and extend the in vitro half-life of the molecule compared to native IGF-1. The three disulfide bonds of native IGF-1 are preserved.

SELECTED PEER-REVIEWED REFERENCES

1. Tomas F.M. et al. - *Anabolic effects of insulin-like growth factor-I (IGF-I) and an IGF-I variant in normal female rats* - Journal of Endocrinology, 1993 - DOI: 10.1677/joe.0.1370413
2. Francis G.L. et al. - *Insulin-like growth factor (IGF)-I and analogues with reduced binding to IGF-binding proteins (IGFBPs): increased mitogenic action and altered IGF-I/IGFBP interactions* - Journal of Molecular Endocrinology, 1992 - PMID: 1378415
3. King R. et al. - *Production and characterization of recombinant insulin-like growth factor-I (IGF-I) and potent analogues of IGF-I, with Gly or Arg substituted for Glu3* - Journal of Molecular Endocrinology, 1992 - PMID: 1378416

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.